

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 8: Grid Navigation (Polar Heading Conversion & Isogrivs)

Syllabus Source: CAE Oxford (pp. 287-324) | Pass Target: $\geq 90\%$ AviationExam

1. WHY GRID NAVIGATION?

Near the geographic poles, magnetic compasses are useless due to vertical magnetic dip, and true meridians converge so rapidly that a straight flight path experiences massive changes in true heading every few minutes:

- The Grid Concept: A rectangular grid is superimposed over the polar chart. All grid vertical lines are drawn parallel to the Greenwich Meridian (000° Longitude). The direction towards the North Pole along the Greenwich Meridian is defined as GRID NORTH.
- Grid Heading / Track Formula (Northern Hemisphere): $\text{Grid Track} = \text{True Track} \pm \text{Longitude}$. Memory rule: In West Longitude, $\text{Grid Track} = \text{True Track} - \text{West Longitude}$. In East Longitude, $\text{Grid Track} = \text{True Track} + \text{East Longitude}$ ('Grid North Is Less In West' $\rightarrow G = T - W$; $G = T + E$).
- Isogrivs & Grivation: Grivation is the angle between Compass/Magnetic North and Grid North: $\text{Grivation} = \text{Variation} \pm \text{Convergence}$. Lines connecting points of equal grivation are ISOGRIVS.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): High-Latitude Grid Track Conversion

Grid Navigation Formula (Greenwich Meridian Datum in Northern Hemisphere): - $\text{Grid Track} = \text{True Track} \pm \text{Grid Convergence}$. - At West Longitude: $\text{Grid Track} = \text{True Track} + \text{West Longitude}$. - At East Longitude: $\text{Grid Track} = \text{True Track} - \text{East Longitude}$. Example: Aircraft at 82°N, 045°W has True Heading 090°. - Since position is West Longitude: $\text{Grid Heading} = 090^\circ + 045^\circ = 135^\circ \text{ Grid}$. - Directional gyros in Free Gyro mode are set to 135° and held constant across meridians.

2. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Converting True Track to Grid Track

SCENARIO (Polar Navigation Route Clearance Drill): An aircraft is flying over northern Canada: - Current position: 75°N 110°W - Aircraft True Track = 060°
True QUESTION: What is the corresponding GRID TRACK to select on the autopilot/FMC? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify the Longitude and Hemisphere: - Longitude = 110° WEST. Step 2: Recall the Grid Conversion Formula for the Northern Hemisphere: - Memory Rule: 'Grid Is Less In West' (Subtract West Longitude!) - Formula: $\text{Grid Track} = \text{True Track} - \text{West Longitude}$ Step 3: Plug in the numbers: - True Track = 060° - Longitude = 110° - $\text{Grid Track} = 060^\circ - 110^\circ = -050^\circ$. Step 4: Normalize negative angles to standard 360° compass format: - Add 360°: $-050^\circ + 360^\circ = 310^\circ \text{ Grid}$! FINAL ANSWER: Grid Track = 310° Grid. Flying 310° Grid will keep the aircraft flying in a perfectly straight line over the pole without touching heading knobs!

[!] EXAM TRAP (AVIATIONEXAM) - The West Longitude Subtraction Trap

- Northern Hemisphere: $\text{Grid} = \text{True} - \text{West Longitude}$ (or $\text{True} + \text{East Longitude}$). - If the result is negative, simply add 360°! - In the Southern Hemisphere, the rule reverses: $\text{Grid} = \text{True} + \text{West Longitude}$.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: On a Polar Grid chart in the Northern Hemisphere, what meridian defines the direction of GRID NORTH?**

- [A]: The 180° Antimeridian
- [B]: The GREENWICH MERIDIAN (000° Longitude)
- [C]: Magnetic North Pole
- [D]: The Equator

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Grid North is defined by lines parallel to the Prime Meridian (000°), pointing towards the North Pole along the Greenwich Meridian.

Q2: In the Northern Hemisphere at position 80°N 040°W, an aircraft is flying on a True Track of 120°. What is the Grid Track?

- [A]: 160°
- [B]: 080° ($\text{Grid} = \text{True} - \text{West Longitude} = 120^\circ - 40^\circ = 080^\circ \text{ Grid}$)
- [C]: 040°
- [D]: 200°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Using the rule 'Grid is Less in West': $\text{Grid Track} = \text{True Track} - \text{West Longitude} = 120^\circ - 40^\circ = 080^\circ \text{ Grid}$.

Q3: What are lines on an aeronautical chart connecting points of EQUAL GRIVATION called?

- [A]: Isogonals
- [B]: ISOGRIVS
- [C]: Isoclines
- [D]: Agonic lines

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Isogrivs are lines of equal grivation (the angular difference between magnetic direction and grid direction).